



III Year B.Tech Application Development III (AD-II) Project Review

EAT-SMART

Batch Details



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Under the guidance of

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Abstract

EAT-SMART is an advanced mobile application designed to support healthy eating habits through intelligent food tracking and weekly dietary analysis. Users log meals from a predefined dataset, and the system evaluates eating patterns to generate structured weekly reports highlighting progress and goal achievement. Based on this analysis, EAT-SMART provides personalized diet tips and motivational encouragement to guide users throughout their dietary journey. Visual charts and progress indicators enable easy understanding of trends and improvements. The application stores data locally using **SQLite** for offline accessibility and is developed using **Flutter and Dart** in **Visual Studio Code**, demonstrating efficient data handling, analytical processing, and user-centric mobile application design.

To develop this project we use algorithms such as :

RULE-BASED CLASSIFICATION, PATTERN ANALYSIS, RECOMMENDATION-ENGINE, PROGRESS SCORING not only that also we use patterns like **TIME-SERIES PATTERN, FREQUENCY PATTERN, GOAL MATCHING PATTERN, BEHAVIORAL INNOVATION.**

Introduction

- Modern lifestyles often lead to **unhealthy eating habits** and irregular meal patterns.
- Maintaining a **balanced diet** is challenging without proper tracking and guidance.
- **EAT-SMART** helps users monitor daily food intake and **analyze weekly dietary patterns**.
- Provides **personalized diet tips** and **motivational encouragement** to stay consistent.
- **Visual progress reports and charts** allow users to track improvement over time.
- Developed using **Flutter and Dart** with **SQLite** for offline data storage.
- Demonstrates **efficient data handling, analytical processing, and user-centric design**.

Existing System

Most current diet apps only track **daily food intake** and calories. They **lack weekly analysis, personalized suggestions, and motivational support**, and often require constant internet access. Limited visual feedback makes them **less effective in helping users maintain healthy eating habits**.

⚠ Limitations of the Existing System

- Only tracks daily food intake, no weekly analysis.
- Lack of **personalized diet suggestions** or motivational support.
- Poor **user engagement**; does not encourage consistency.
- Often **internet-dependent**, restricting offline usage.
- Visual reports and progress tracking can be **confusing or inadequate**.
- Limited ability to **analyze eating patterns** over time.

Proposed System

EAT-SMART provides **weekly analysis, personalized diet tips, and motivational support** to help users maintain healthy eating habits. It generates **visual progress reports**, tracks meals, and offers guidance for consistency, all while being **offline-accessible and user-friendly**.

★ Key Features of EAT-SMART:

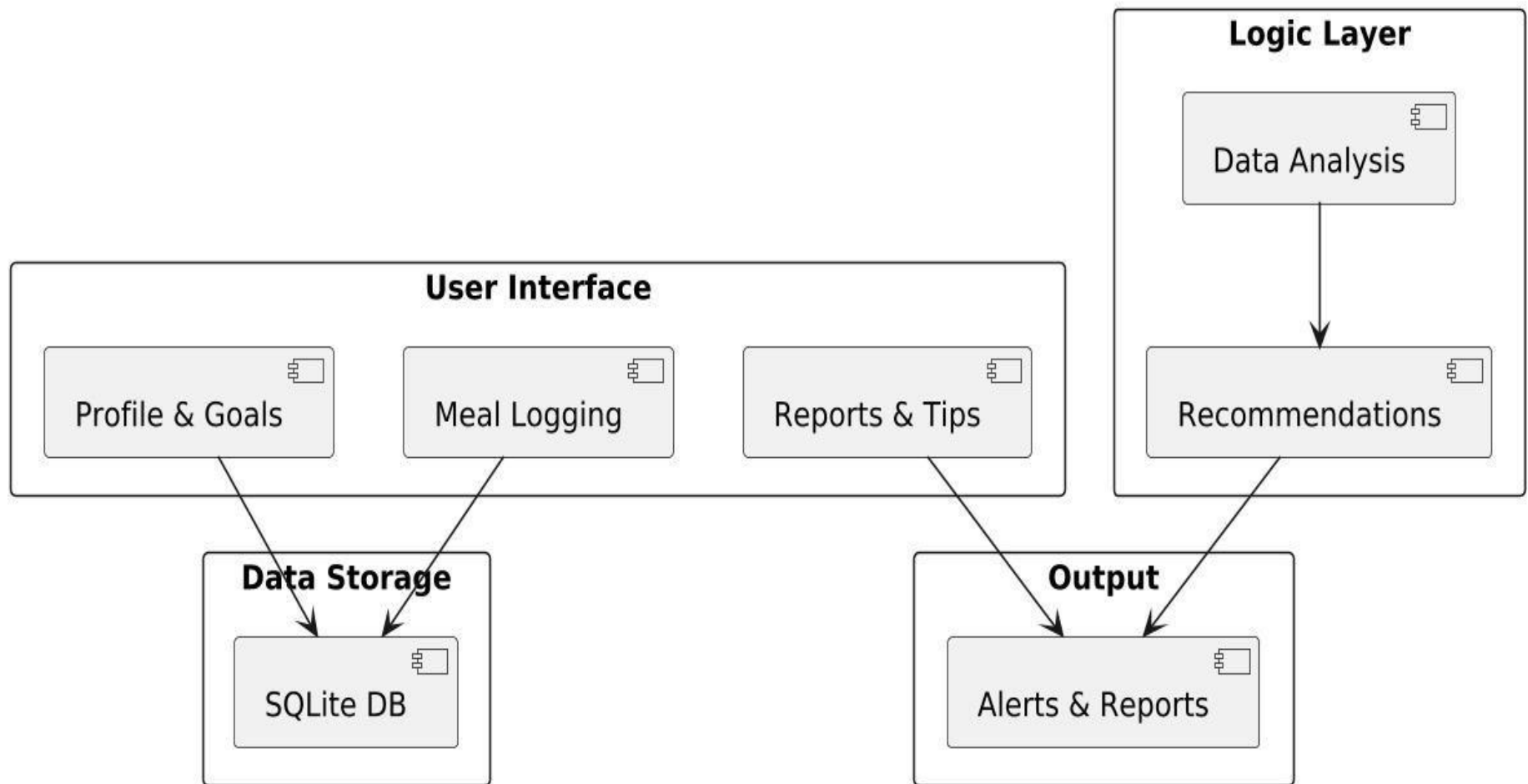
- Tracks **daily food intake** and **weekly patterns**.
- Provides **personalized diet suggestions** based on user goals.
- Generates **visual progress reports** and charts.
- Sends **motivational messages and encouragement**.
- Works **offline** using local SQLite storage.
- Designed to be **scalable for AI or cloud integration** in future.
- It also **provides reviews/feedbacks** from the previous users.



System Architecture

System Architecture

EAT-SMART System Architecture





Technical Requirements Specification (TRS)

Software Requirements Specification(SRS)

- **Framework & Language:** Flutter & Dart
- **IDE:** Visual Studio Code
- **Database:** SQLite (local storage)
- **UI:** Login, Meal Logging, Reports & Tips (Flutter widgets)
- **Logic:** Data Analysis & Personalized Recommendations
- **Features:** Weekly charts, motivational alerts, offline access
- **Libraries:** sqflite, flowchart, provider/bloc, flutter_local_notifications

Hardware Requirements Specification(HRS)

PC/Laptop: Intel i3/i5, 8 GB RAM, 256 GB storage

Mobile Device: Android 7+, 2 GB RAM, 16 GB storage

OS: Windows/macOS/Linux

Connectivity: USB or wireless for testing

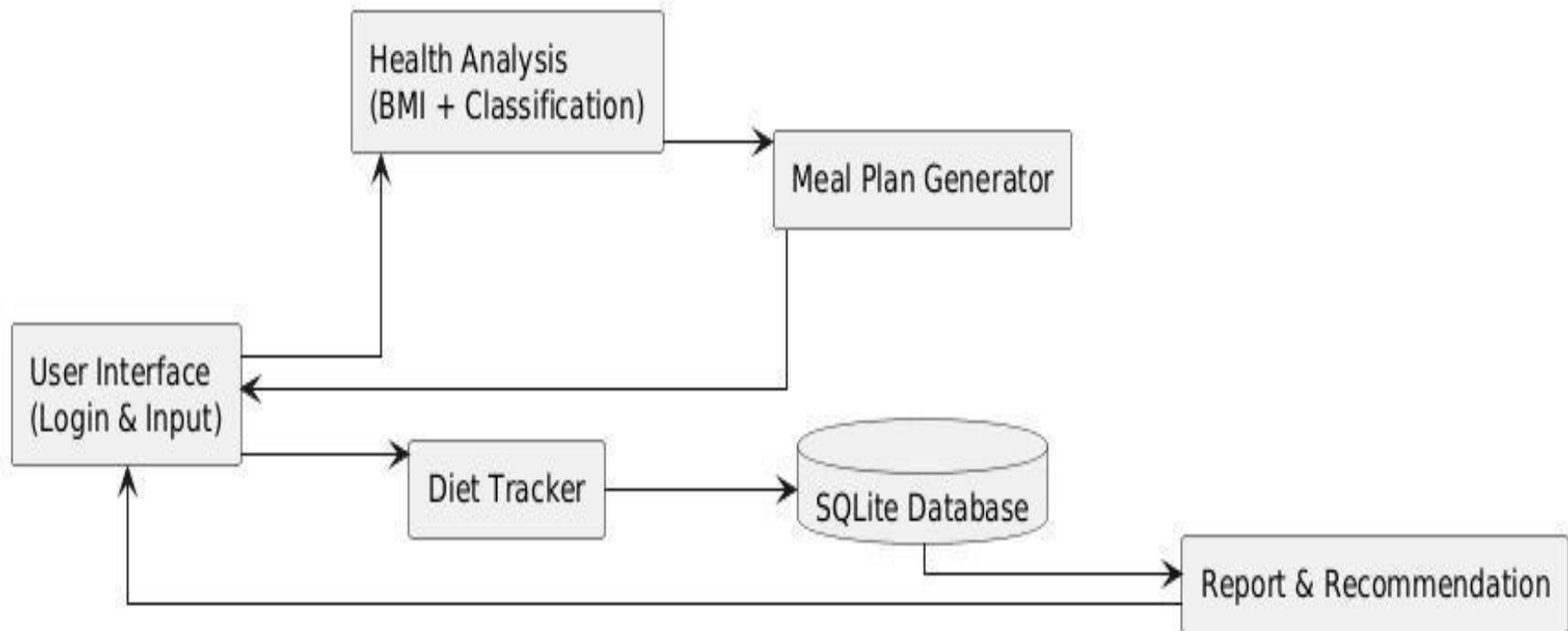
Application Modules

1. User Authentication Module
2. User Profile Management
3. BMI Calculation and Classification Module
4. Meal Plan Generator
5. Daily Diet Tracker
6. Pattern Analysis Module
7. Recommendation Engine
8. Report Generation Module
9. Local Data Storage Module
10. User Interface Module

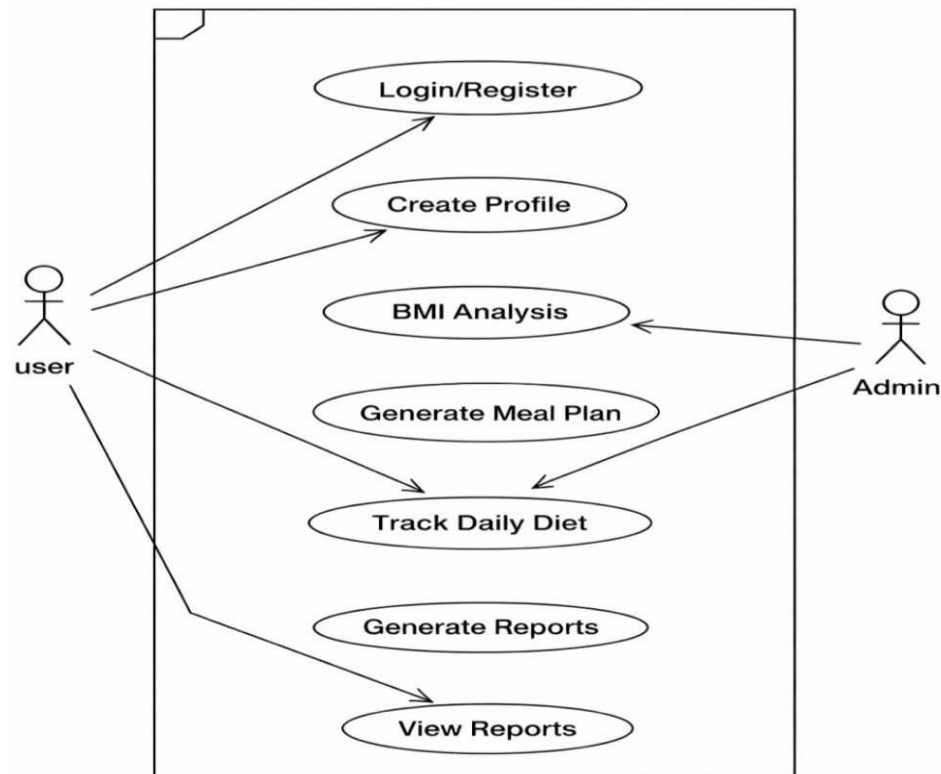
System Models (UML Diagrams)

Dataflow Diagram

EAT-SMART Data Flow Diagram

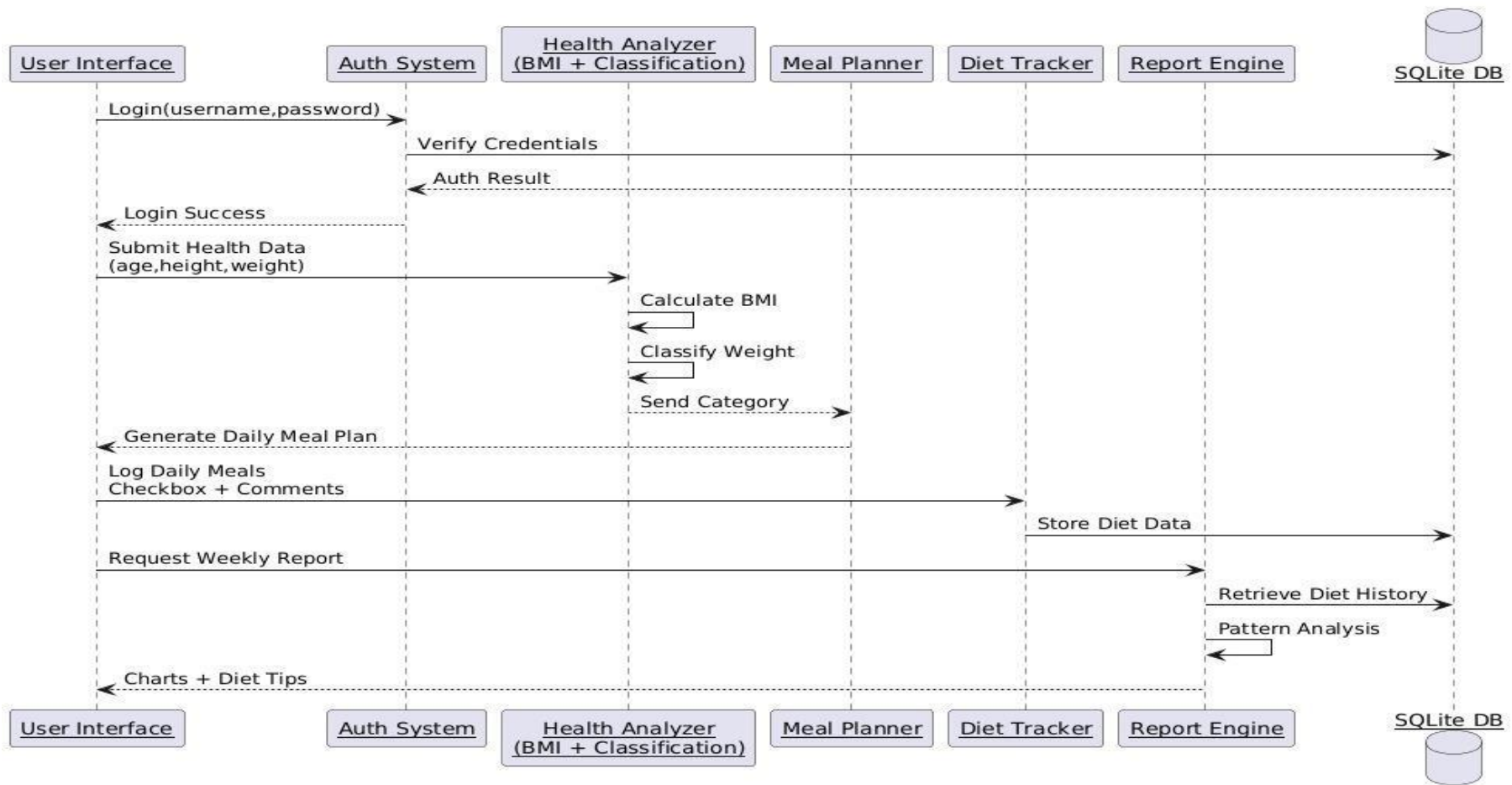


Use Case Diagram

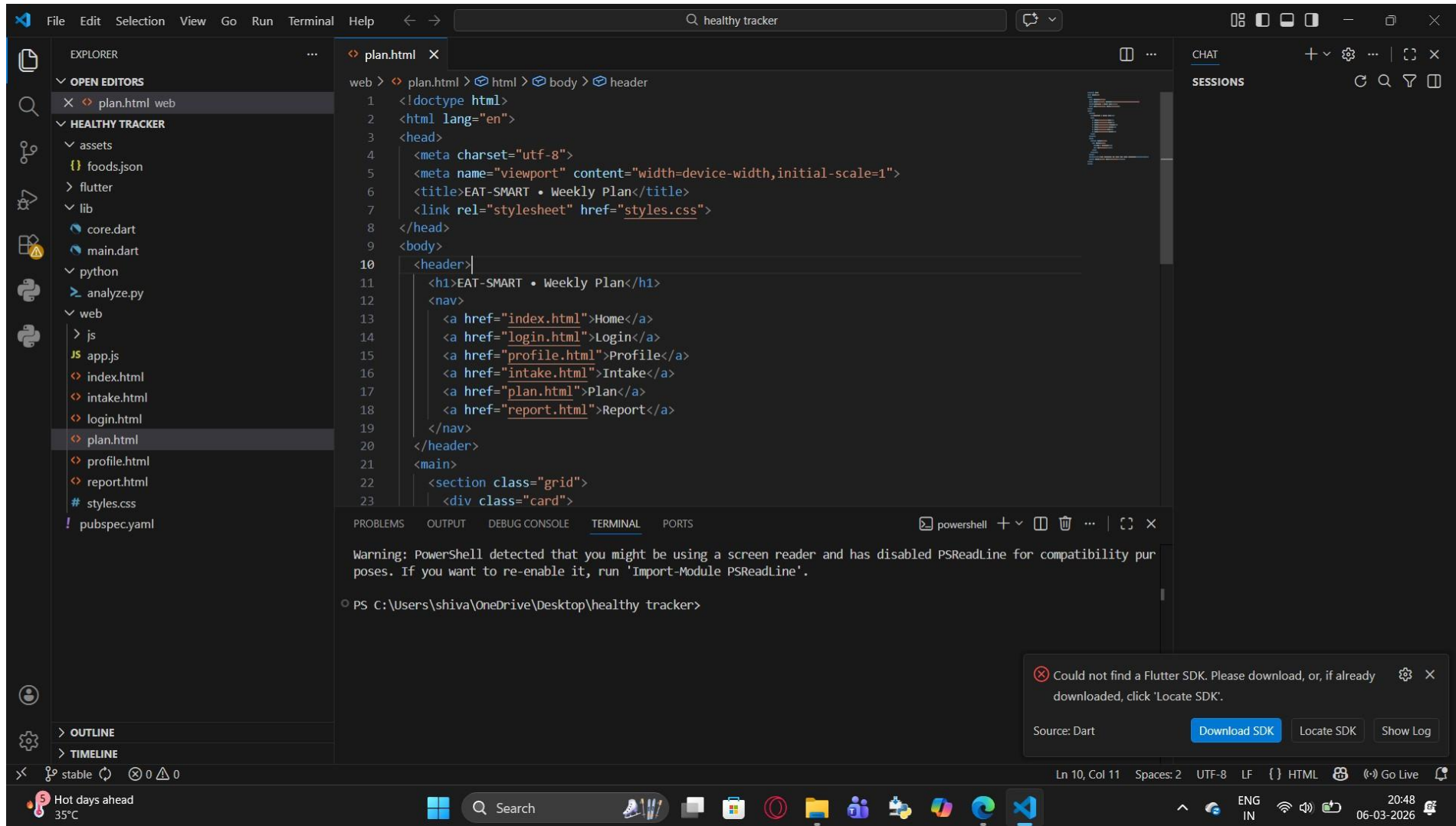


Sequential Diagram

EAT-SMART Sequence Diagram



Implementation



```
1 <!doctype html>
2 <html lang="en">
3 <head>
4   <meta charset="utf-8">
5   <meta name="viewport" content="width=device-width,initial-scale=1">
6   <title>EAT-SMART • Weekly Plan</title>
7   <link rel="stylesheet" href="styles.css">
8 </head>
9 <body>
10  <header>
11    <h1>EAT-SMART • Weekly Plan</h1>
12    <nav>
13      <a href="index.html">Home</a>
14      <a href="login.html">Login</a>
15      <a href="profile.html">Profile</a>
16      <a href="intake.html">Intake</a>
17      <a href="plan.html">Plan</a>
18      <a href="report.html">Report</a>
19    </nav>
20  </header>
21  <main>
22    <section class="grid">
23      <div class="card">
```

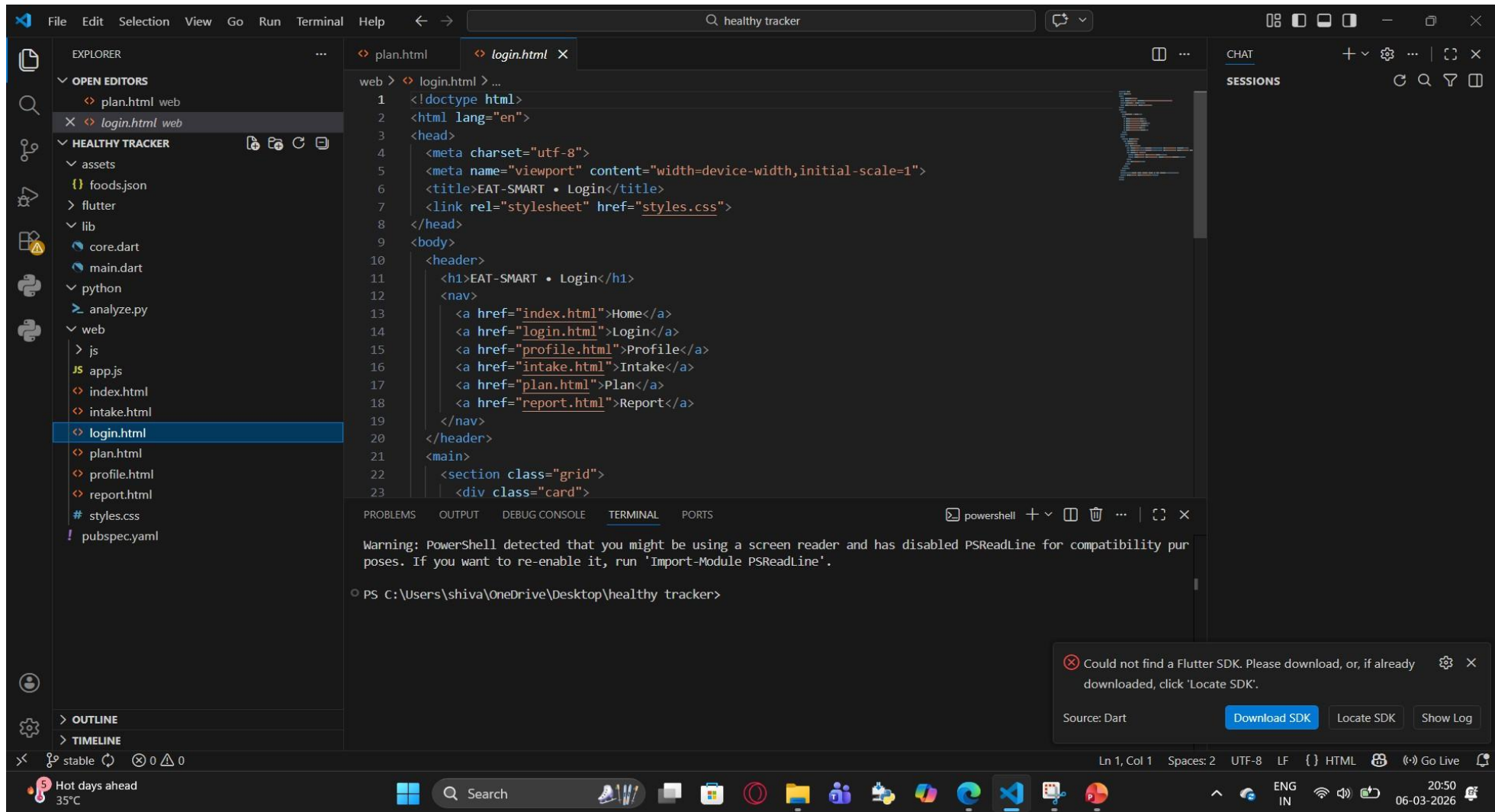
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\shiva\OneDrive\Desktop\healthy tracker>

Could not find a Flutter SDK. Please download, or, if already downloaded, click 'Locate SDK'.

Source: Dart [Download SDK] [Locate SDK] [Show Log]

Implementation



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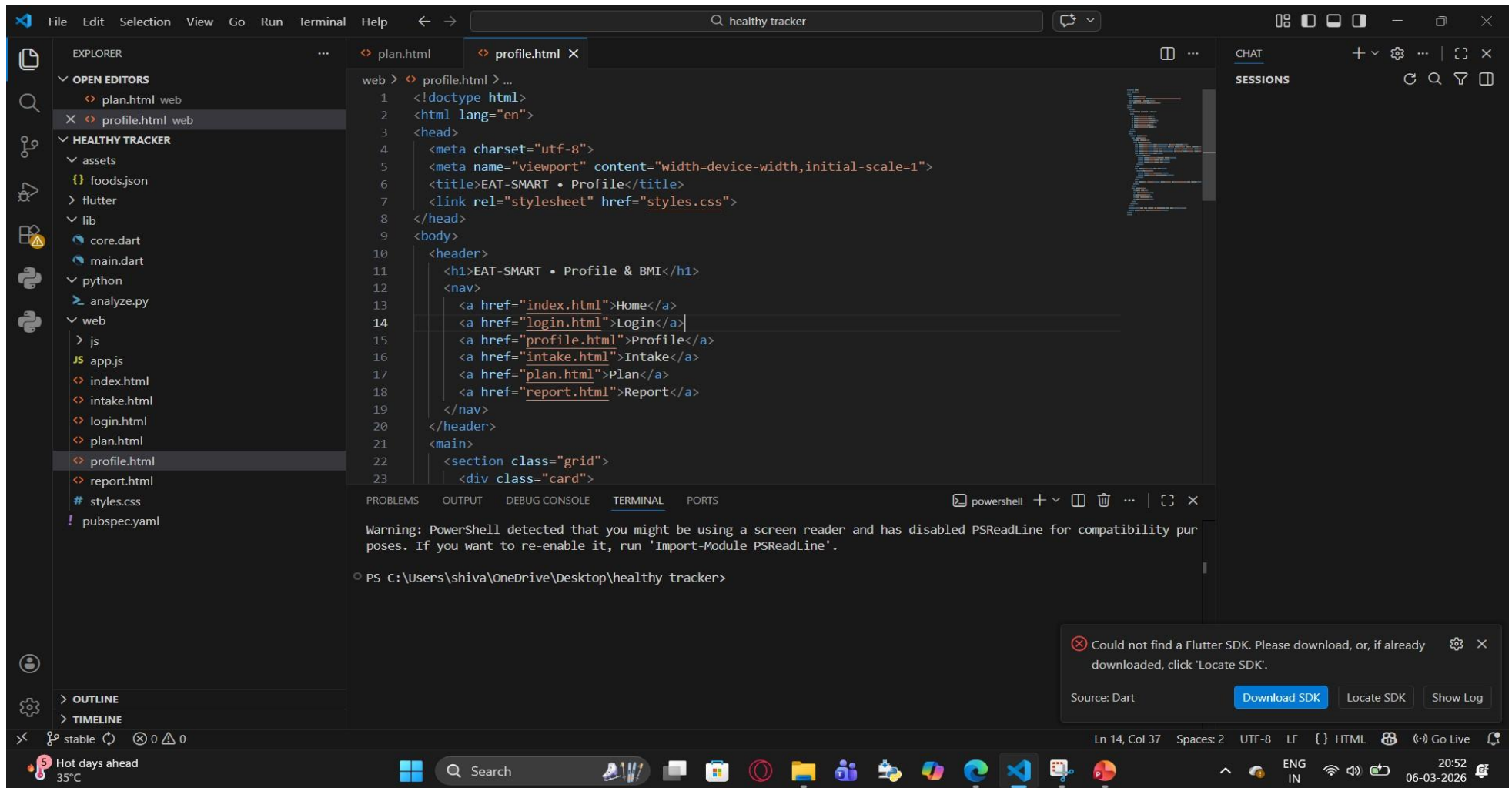
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Implementation



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6   <title>EAT-SMART • Profile</title>
7   <link rel="stylesheet" href="styles.css">
8 </head>
9 <body>
10  <header>
11    <h1>EAT-SMART • Profile & BMI</h1>
12    <nav>
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14      <a href="login.html">Login</a>
15      <a href="profile.html">Profile</a>
16      <a href="intake.html">Intake</a>
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Source: Dart [Download SDK] [Locate SDK] [Show Log]

Ln 14, Col 37 Spaces: 2 UTF-8 LF {} HTML Go Live



Output Screens



III Year B.Tech Application Development II (AD-II) Project Review

EAT-SMART

File C:/Users/shiva/OneDrive/Desktop/healthy%20tracker/web/index.html

☆ ⚙ ⋮ 👤 Chat

⌵ 🗑 📁 New folder 🔗 My Next Step - Cr... 📄 New tab 🌐 GDG on Campus H... 🌐 Google Developer... 🛡 McAfee Security

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EAT-SMART

Professional multi-page experience

Login Profile Intake Plan Report

Welcome

Track your nutrition, analyze BMI and weekly trends, and follow tailored daywise plans.

Get Started Set Profile

Features

- Login/Register and secure local account
- BMI analysis and category-based tips
- Daily intake logging with calorie calculation
- Daywise plan with checkboxes and comments
- Weekly report chart and motivation

User-friendly AI-based tracker with weekly reports & plans

🔔 Hot days ahead 35°C

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III Year B.Tech Application Development II (AD-II) Project Review

File C:/Users/shiva/OneDrive/Desktop/healthy%20tracker/web/login.html

Home Login Profile Intake Plan Report

EAT-SMART • Login

Account

Username

Password

[Login](#) [Register](#)

Secure local account stored in your browser



EAT-SMART • Profile & BMI

Home Login Profile Intake Plan Report

Your Details

Name

pranavi

Age

21

Weight (kg)

65

Height (cm)

158

Goal

Weight Loss

Diet Type

Vegetarian

BMI & Tips

BMI 26.0 • Overweight

- Reduce refined carbs
- Add vegetables and lean protein
- Avoid junk food
- Drink more water

Diet Suggestions

- Prioritize vegetables, lean protein, whole grains
- Reduce refined carbs and avoid junk food
- Target modest calorie deficit and daily activity
- Drink more water

Log Food

Day

Mon

Food

Rice (130 kcal)

Quantity

1

Add to Day

Go to Plan

Health Conditions

Condition

e.g., fever, accident

Add Condition

- healthy

Plan & Checklist

Mon • Done



- Roti x1
- Paneer x1

oh wow i feel lighter

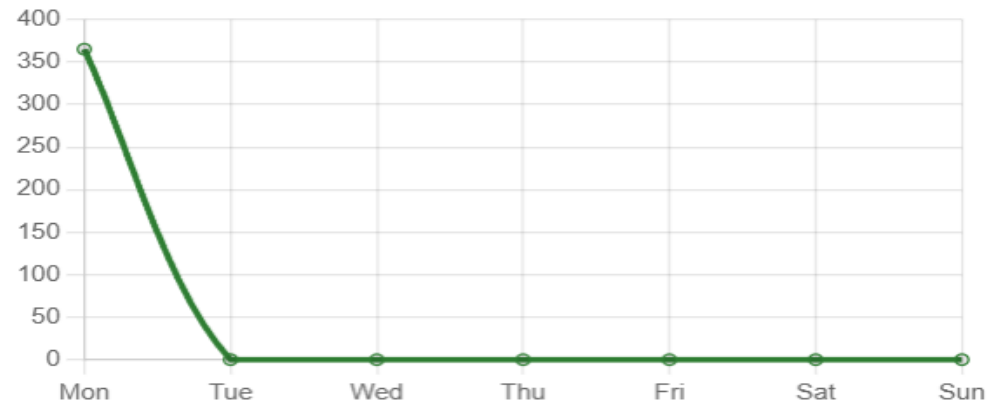
Tue • Pending



- Breakfast: Oats + Yogurt
- Lunch: Dal + Brown Rice
- Dinner: Salad + Omelette
- Snack: Fruit

Your experience for the day

Calories Trend



Insights

Tips

- You are consuming fewer calories than required
- Drink more water
- Avoid junk food

Motivation

- Consistent logging
- Keep improving
- Stay hydrated

Conclusion & Future Scope

Conclusion

EAT-SMART is an intelligent mobile application that helps users maintain healthy eating habits through effective food tracking and analysis. It provides personalized suggestions, visual progress reports, and motivational support to guide users toward their dietary goals.

- Tracks daily meals and analyzes weekly patterns
- Provides personalized diet recommendations
- Displays progress through visual charts and reports
- Encourages users with motivational feedback
- Works offline using SQLite for reliability
- Scalable for future AI and cloud integration

Future Scope

EAT-SMART has strong potential for future enhancements by integrating advanced technologies and expanding its features. With improvements like AI-based recommendations and cloud connectivity, the application can provide more accurate insights and a better user experience.

- Integrate AI/ML for smarter and personalized diet suggestions
- Enable cloud storage for data backup and multi-device access
- Add wearable device integration for real-time health tracking
- Implement barcode scanning and image-based food recognition
- Expand the food database with real-time updates
- Introduce social features like challenges and community support
- Provide multilingual support for wider accessibility
- Offer advanced health analytics and insights

Thank You

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